

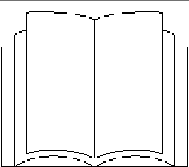


Operator's manual for Attachment



Winch

Product number:	300-700-series loaders:	A33130
	200-series loaders:	A33151



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

AVANT[®]
AVANT TECNO OY
e-mail: sales@avanttecno.com

Ylötie 1
33470 Ylöjärvi
FINLAND
Tel. +358 3 347 8800
Fax +358 3 348 5511

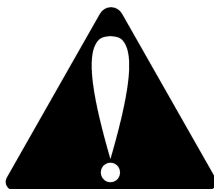
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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or damage to property. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This operator's manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, even an person with only a little experience can use the attachment and loader safely. The manual also includes important instructions for experienced AVANT operators. Ensure that all persons using the equipment have received proper guidance and familiarised themselves with the manual and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The AVANT winch attachment is designed and manufactured to be suitable for use with AVANT multi purpose loaders. The winch is suited, for example, to pull tree trunks, and objects that cannot be reached with the loader. The winch can also used to pull the loader itself in case it has got stuck. Normally the winch is mounted on the quick attach plate on the loader boom. Alternatively, with the rear mounting bracket or the rear lift device, available on some loader models, the winch can be mounted to the rear of the loader, which leaves the boom free for some other attachment. The winch is driven directly hydraulically. It is equipped with a safety net in case the wire snaps, 30 m of wire rope and a hook with safety latch.

Mounting the winch on the loader boom makes it possible to lift the towed object simultaneously. The winches are not designed to be used on solely lifting purposes, as the winch is not equipped with safety features needed in lifting operations. The winch should not be left loaded, or for example to support other structures.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. Not all maintenance tasks can be performed by the operator; certain tasks must be left to qualified professionals. Appropriate protective equipment and original spare parts must be used for the maintenance. Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the winch

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards.

The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- monitor and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- When mounting the winch to the loader, ensure that the locking pins of the loader's quick attach plate are in the lower position and that they **have locked the attachment to the loader**.
- Operate the winch carefully and calmly.
- The operating area must be suitable for winch operation and adequately illuminated.



- Keep a safety distance of at least 2 meters while the winch is operating. Handle the wire rope with care especially when reeling the rope.
- The winch is to be used by one operator at a time. Do not operate the winch controls if another person is handling the wire rope.
- The wire rope must be checked regularly and at least annually. If broken or bent threads are discovered the entire wire rope must be replaced immediately. Read the instructions for inspections, maintenance and service.
- Always transport the load as low and as close to the machine as possible to keep the centre of gravity low.
- Never go under a raised attachment or the loader boom.
- Pay attention to the contours of the terrain and the hazards related to uneven ground.
- Make sure the loader is properly supported before using the winch. Do not use the winch on inclined terrains. Avoid pulling from the sides with the winch.
- Pay attention to the surroundings and any other persons and machines moving in the vicinity. Move carefully and calmly near the animals.
- Shut down the loader engine before any cleaning, maintenance or adjustments.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Also read the safety instructions for the loader in the loader manual.



Note that the loader boom moves down even if the engine has been stopped. Never go under a raised boom or the attachment. Also note that if the winch is left pulling a load, the pulling force will gradually fade when the engine has been shut.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.

Remember to wear proper protective clothing:



- The noise level in the driver's seat exceeds 85 dB(A). Wear hearing protection while working with the loader.



- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic hoses.



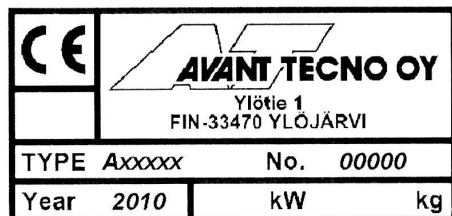
Never wear loose clothing, long scarves, jewellery or other items (such as a wrist watch, rings) that might become entangled with the wire rope. If needed, also wear a hat to prevent long hair from coming into contact with the wire rope or the winch.

4. Winch specifications

Listed on the following page are the markings and labels fastened to the attachment, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover page.



The warning decals contain important safety information. Replace damaged or missing decals with new ones.



Identification plate of the attachment

- A46771 – Read instructions before use.
- A46772 – Do not go under a raised attachment, stay away from raised equipment.
- A46801 – Entanglement hazard. Keep a safe distance. Keep clear of moving parts.



A46771



A46772



A46801

Table 1 – Winch technical data

Product number	A33151	A33130
Suitable AVANT loader*	200 series	300 - 700 –series
Winch length / width / height:	810 mm / 1 100 mm / 700 mm	
Wire rope:	30 m, steel	
Weight, winch frame:	50 kg	
Total weight with 30 m rope:	105 kg	
Pulling force:	1000 kp*	
Hydraulic motor:	315 cc	
Maximum input of hydraulic energy:	22,5 MPa (225 bar), 70 l/min	

*Actual pulling force depends on the loader model and the amount of wire rope on the drum.

5. Assembly of the winch

The winch is delivered in working condition. Mount the attachment and its hydraulic hoses according to the following instructions and practise its use in a safe area. Read the following instructions for mounting, using and maintenance of the equipment before putting it into service.

5.1. Mounting the winch to the loader

Mounting the attachment to the loader is quick and easy, but it must be done carefully. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3.

The winch may be mounted to the rear of some specific loader models. For rear mounting, the optional double acting rear auxiliary hydraulic outlet is required, as well as the rear lifting device or a specific winch mounting bracket. Ask your Avant dealer for more information.

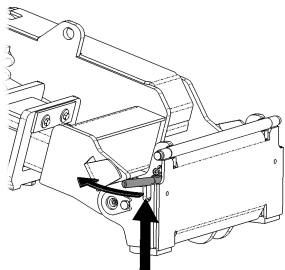
If your loader is equipped with a hydraulic locking and/or serial quick attach system, refer to the relevant manuals for additional mounting instructions.

The attachment is mounted to the AVANT loader boom quick attach plate and the counterpart on the attachment. The locking is semi-automatic; the locking latches are released as the locking pins hit the pin guides when the boom front plate is tilted towards the loader (300-700-series) or the locking pin hits the boom (200-series). The attachment is mounted to the loader as follows:

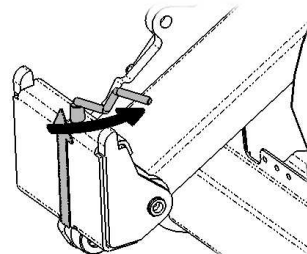


Never move or raise an attachment that has not been locked.

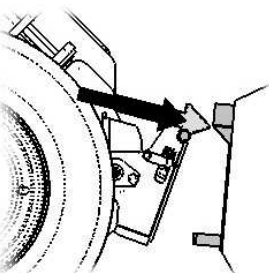
Step 1:



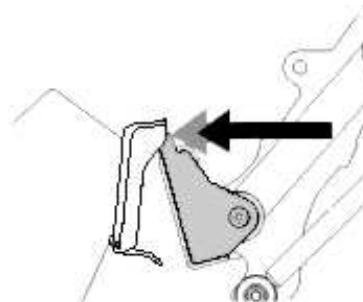
- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- Ensure that the hydraulic hoses are not in the way during installation.



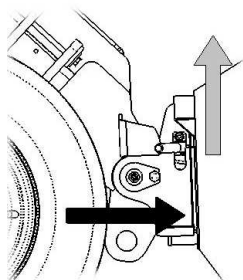
Step 2:



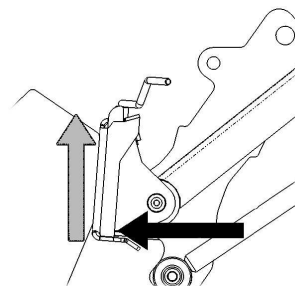
- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.



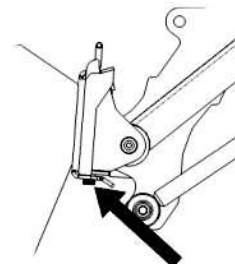
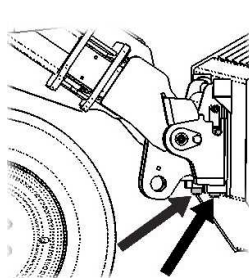
Step 3:



- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Turn the boom control lever further to the left so that the locking pins lock the attachment as they hit the pin guides.



Before moving or raising the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides. Unlocked attachment may fall on the boom or operator and cause injuries.



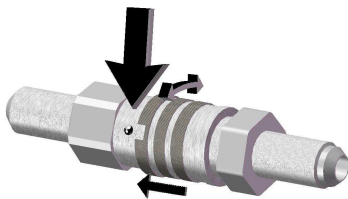
Do not use the automatic locking of the locking pins **at a height of over one metre**. If the locking pins do not return to the normal position when the front plate is tilted, do not raise the attachment any more. Lower the attachment on the ground and secure the locking manually.

5.2. Connecting and disconnecting the hydraulic fittings

The winch has three hydraulic fittings, which must be connected to the corresponding fittings on the loader. The hoses should be connected so that the fittings with coloured caps are connected. The remaining two pairs will be then connected correctly.

To connect and disconnect the fittings, move the collar at the end of the female fitting. When connecting or disconnecting fittings, shut down the loader engine, move the auxiliary hydraulics lever to its extreme positions a few times. This way, will not be residual pressure left in the system and the fittings are easy to connect.

In a 200-series machine, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn by accident into the locking position, making it impossible to connect or disconnect the fittings.



In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled downwards to disconnect the fittings from each other.



Always ensure that all three fittings are properly connected to the corresponding fittings on the loader. If any of the hoses is not properly connected, the hydraulic motors of the snow blower may be damaged.



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.



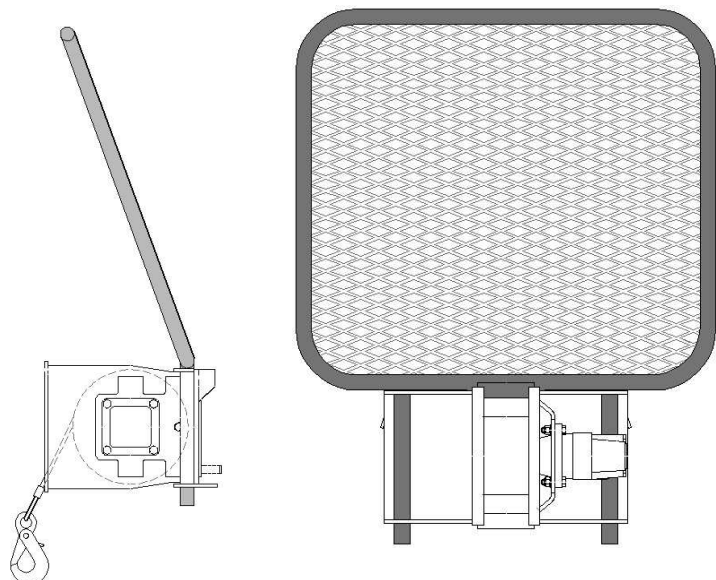
Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure. High-pressure fluid ejection may cause injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. Never leave the hoses hanging on the ground; place them on top of the equipment.

5.3. Mounting the safety net

With the winch, a safety net is provided, which is to catch the wire rope that has come loose or broken, preventing it from reaching the operator.

Slide the safety net to the winch so that it is tilted forward, as shown in the adjacent figures.



6. Instructions for use

Work can be started when the winch has been mounted and the hydraulic fittings connected according to the instructions. Check the attachment once more before starting work. Checking of the winch, the wire rope and the hook are a part of ensuring safe operation.

6.1. Pre-use checks

- Check that the safety net is in place.
- Check the condition of the wire rope and the hook. See section 7.2 for more information.
- Make sure that the winch is operated to the correct rotating direction. See the safety message below.
- Check regularly to ensure there are no hydraulic leaks. Refer to Section 7 for instructions.
- Check that all obstacles, have been removed from the way of the towed object. Remember correct working methods and leaving the driver's seat unnecessarily.
- Ensure that the loader and attachment are used in a safe manner and as intended. Do not allow children to operate the machine.
- Operate the attachment only from the driver's seat.



Do not operate the winch, when it has been connected so that the auxiliary hydraulics lever of the loader will lock the winch to the pulling operation. If the winch rotates in a way that it winds the wire rope when the auxiliary hydraulics lever is actuated towards its locking position, the hoses of the winch are connected the wrong way. Change the coupling order of the male fittings.



Before using the winch check that the wire rope, hook, their fastening, and all the additional equipment used, are intact. Only use wire rope or hook that is in good condition and similar to the original ones.

6.2. Operation

After the winch has been connected, check the rotating direction of the winch. To make it possible to pull cable from the reel, the wire rope should get unreeled when the auxiliary hydraulics lever is actuated towards its locking position.

1. Drive the loader so that the winch will pull the object from a straight direction.
2. **Lock the parking brake of the loader.**
3. Set low engine revolutions and lock the auxiliary hydraulics lever to unreel the wire rope. Unreel a suitable length of the wire rope and stop the winch.
4. Fix the hook properly to the object being pulled. Avoid damage to the wire rope by using suitable additional chains or straps. Do not tie the wire rope, otherwise it will get damaged or may get loose suddenly.



Always wear thick leather gloves when handling the wire rope. Do not handle the wire with bare hands; slipping of the wire or a damaged wire rope may cause injuries.



Do not unwind the reel completely. Leave at least five rounds of wire on the reel, so that the wire fixing point will not get overstressed.

Make sure, that the hitching point is secure enough. A wire rope that comes loose may cause a hazard regardless of the safety net. Be especially careful when using the winch to tow the loader stuck in the terrain. The pulling force of the winch is the highest when there is only a little wire left on the reel. In this case the pulling speed is at the lowest.

Operate the winch from the driver's seat only and remember to apply the parking brake. Use low engine revolutions and one auxiliary hydraulics pump. Before starting to tow the object, tighten the rope carefully by reversing the loader or by operating the winch. Lifting the boom during pulling helps to prevent the object being towed from getting stuck. Do not lift the boom so that sticking of the towed object could cause the loader to lose its balance. Stop the pulling in good time; stop towing at the latest when the

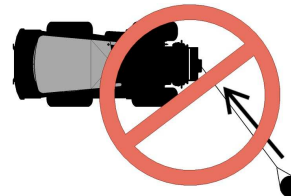


Ensure that there are no other persons at the operating area or near the wire rope. When starting to reel a loose rope, a whip effect may result as the wire rope tightens. The jerk may loosen the fixing of the wire rope or it may be hazardous to persons near the equipment or the rope. Tighten the wire rope slowly before starting to pull the load.



The pulling must be stopped at the latest when the towed object is at three meters from the winch. Do not drive too close to the object to be towed, leave enough space for working around the loader.

Pull the load from directly ahead of the winch. Pulling sideways may cause the wire rope to come off from the reel. Pulling from the sides will also affect the stability of the loader.



Stop pulling with the winch immediately if you notice that the wire rope tends to form a loop or a tight turn, or if the rope is against a sharp corner.

A loose wire rope must be reeled in when a load is applied. A rope that is loose may become entangled and it will not reel evenly, and this might lead to damage to the wire rope. Stop the winch immediately if you notice that the wire rope is not reeling evenly. Unwind some wire rope, if necessary, and start again.

When not using the winch, slide the towing hitch to the corner of the winch. Do not pull the hook tightly in or against the winch, so that the wire rope or the hook will not get damaged. Do not reel more than 30 m of wire rope on the reel.

6.2.1. Operation on uneven terrains

Driving on inclined or uneven terrains and slopes requires extra caution. Watch for holes, bumps, rocks and other possible hidden obstacles. Uneven terrain could overturn the machine. Use low speed and do not make sudden changes in speed or direction.

Try driving straight up or down the hill, not across, to avoid excessive side tilting. Exercise extreme caution when driving near ditches or embankments as the machine could suddenly turn over if an edge caves in. Make sure that the loader is on a surface that is firm enough for towing with the winch, so that the pulling of a load will not result in tipping over of the loader.

7. Inspections, maintenance and servicing

Continuous maintenance includes regular cleaning, as well as monitoring the condition of the wire rope and the hook.



Do not perform any maintenance or service while the loader boom is lifted. The boom may lower by itself and cause serious injuries. Uncouple the hydraulic fittings for maintenance.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Spare parts are available from your nearest AVANT retailer or service point. Contact service if unsure how to perform the repairment safely.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



7.1. Cleaning, lubricating and storing the equipment

Clean and thoroughly dry the wire in case it has become dirty. To guard the rope, use thin oil that will adhere in the rope. Do not grease the rope. Thick grease will speed up the accumulation of dirt into the wire rope. Dirt will have an adverse effect on the wire and its service life.

Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example. Protect the winch from rain.

7.2. The wire rope and the hook

When checking the condition of the wire rope, pay attention to possible cut threads and irregularities on the rope. In a rope that has bent sharply, the threads are applied to uneven loads, and the load will stretch the threads that are under the highest loads. It is not possible to recover a bent rope and a permanently bent rope must be replaced.

Check the hook and its safety latch. Regularly check the fixing of the hook on the rope. Replace the wire rope and the hook with similar as the originals. Do not use damaged equipment.

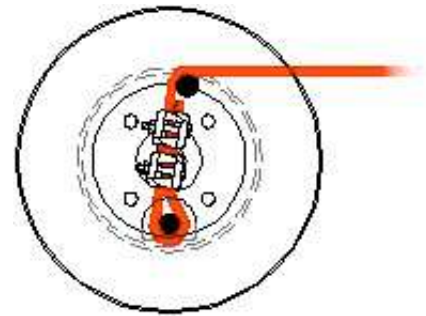
If the wire rope gets stuck on the reel, you can use the loader as a help to unwind the rope. Always make sure, that there are no loops or knots, or that the pulling will not cause bending of the wire rope. Check the wire after clearing the jamming.

7.3. Fastening of the wire rope

The wire rope is fixed on the reel with two shackles. Sufficient tightness of the shackles is important, so that the wire rope will not come off from the reel during pulling. To lower the stress applied on the shackles do not unwind the reel completely, but leave at least five rounds of rope on the reel.

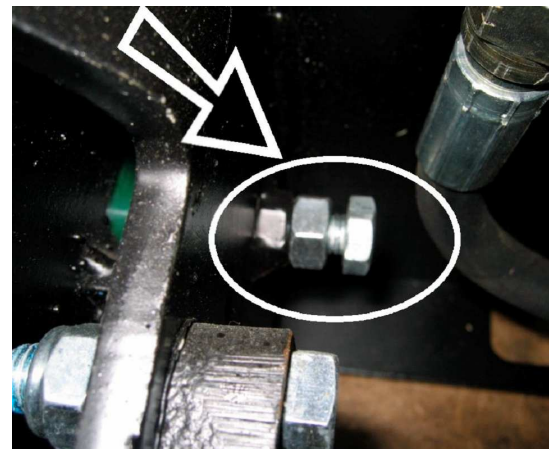
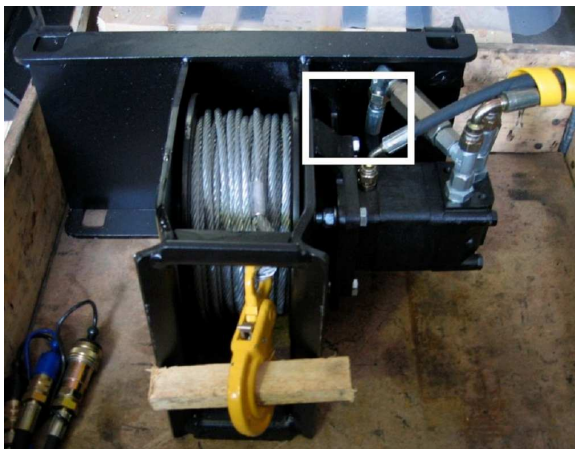
Check the tightness of the shackles at least annually or every 20 hours of use. The shackles, shown in the adjacent figures, can be tightened from the right side of the winch.

When replacing the wire rope, fix the new in the same way as the original. Note the correct turning direction on the reel, the wire should lean against the block fixed on the reel. The wire is coiled on the reel from the upper side. The correct winding direction is counter-clockwise in the adjacent figures.



7.4. Balancing the reel

To balance the rotation of the reel, there is a slide pad next to it. When the winch is operated, high loads are applied to it, and the reel tends to turn. The slide pad should be adjusted close to the reel side or to touch the side of the reel lightly; to keep the reel as straight as possible.



8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) for the attachment from the date of purchase.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work, such as the replacement of blades or the belts, or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:
Osoite / Adress / Address / Adresse:

AVANT TECNO OY
 Ylötie 1
 33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100-1 -2, SFS-EN ISO 14121-1, SFS-EN 982, SFS-EN 14492-1

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100-1 -2, SFS-EN ISO 14121-1, SFS-EN 982, SFS-EN 14492-1

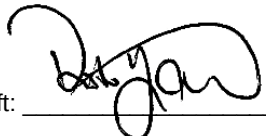
We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100-1 -2, SFS-EN ISO 14121-1, SFS-EN 982, SFS-EN 14492-1

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:
 SFS-EN ISO 12100-1 -2, SFS-EN ISO 14121-1, SFS-EN 982, SFS-EN 14492-1

Kuvaus: Avant-kuormainten kanssa käytettävä hydraulitoiminen vinssi
 Beskrivning: Hydraulisk vinsch; arbetsredskap för Avant lastare
 Description: Hydraulic winch; attachment for Avant loaders
 Beschreibung: Hydraulische Seilwinde; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant	200 series	300-700 series
Vinssi Vinsch Winch Seilwinde	A33151	A33130

Allekirjoitus / Namnteckning / Signature / Unterschrift:  1.7.2010 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





AVANT[®]